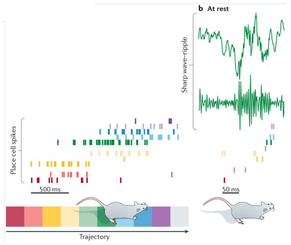
Cortical dynamics in PPC and S1 during awake rest and HPC ripples

Cass C. Delvira Vandevoorde[1], Wing (Winny) Ning[2], Bruce L. McNaughton[2]

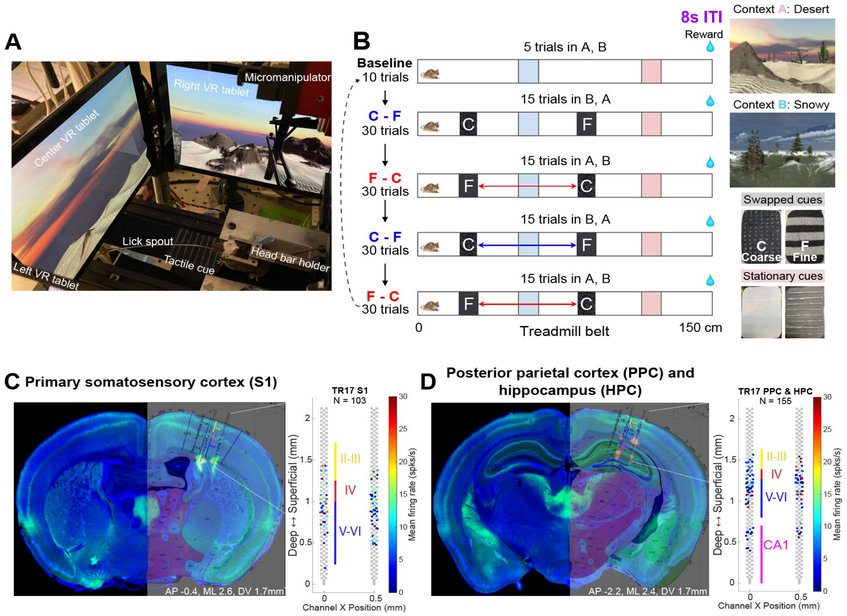
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Introduction

- Hippocampal (HPC) ripples during rest are associated with cortical reactivation of memories to support memory consolidation
- Are there regional and temporal differences between **context** (ex. VR environment) vs. **content** (ex. tactile cue) reactivation during awake rest?

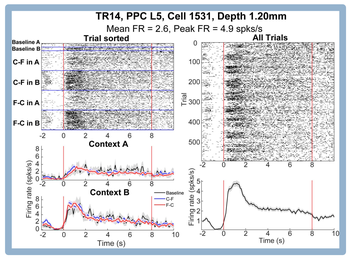


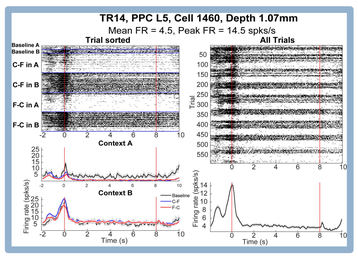
VR Treadmill Task

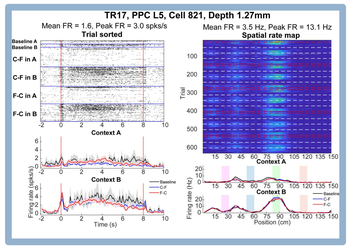


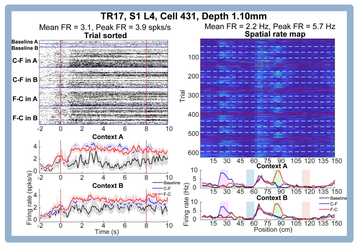
A) Treadmill with VR setup for head-fixed mice. **B)** Behavioral task design. At the end of each trial, the mouse receives a reward followed by an 8 sec inter-trial interval (ITI) when the brake is on and VR tablets are black. **C)** Histology showing the 256F silicon probe recording locations (DiD dye) in S1 and in **D)** PPC and dorsal CA1, stained with DAPI, WFS1 (L2), and NECAB1 (L4) antibodies. Probe map shows the single unit yield across depth in each region, color coded by the mean firing rate.

Single cell activity during awake rest

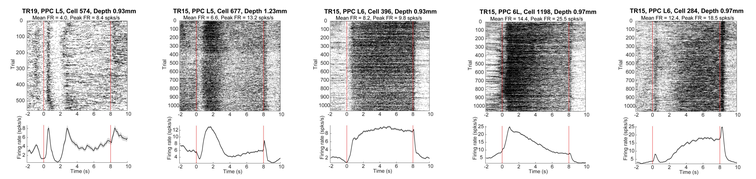




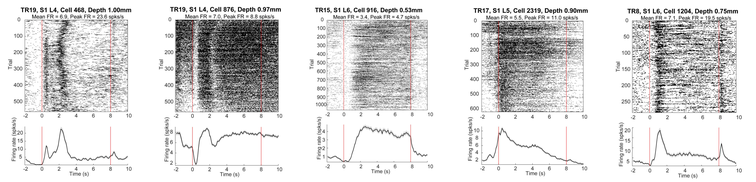




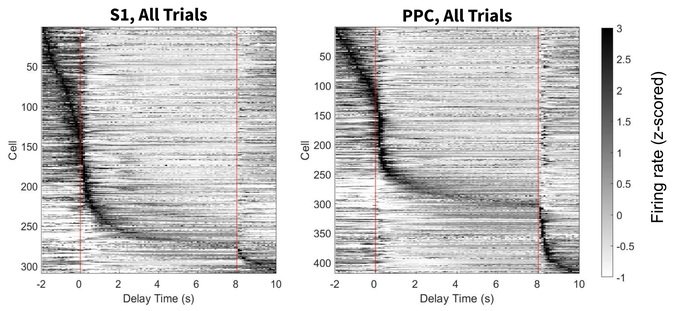
Posterior parietal cortex (PPC)



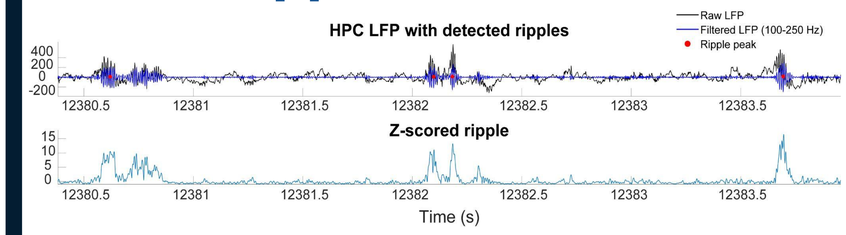
Primary somatosensory cortex (S1)



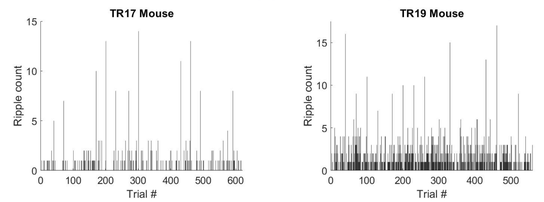
Population activity during 8s ITI (N=6 mice)



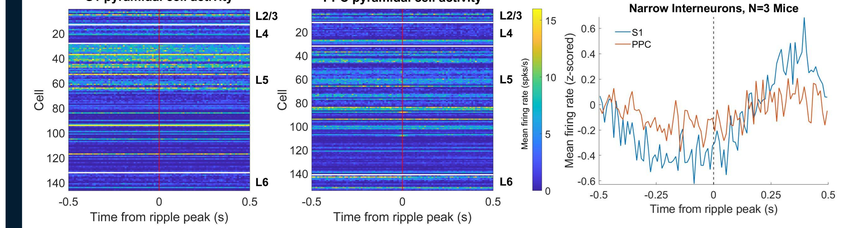
Ripple Detection



Number of ripple events per ITI



Cortical activity around HPC SWR



Results

- We analyzed cortical spiking activity during the 8 sec inter-trial intervals and found some **context-selective cells** in PPC, despite the absence of external, visual stimuli
- Many PPC and S1 neurons showed **ramping or decaying** activity during the intertrial interval
- S1 narrow interneurons show peak activity ~0.35 sec after HPC ripple peak times

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